

An LLM-Based Clinical Decision Support System

Abstract

Clinical documentation contains critical patient data, but its unstructured nature makes it difficult for practitioners to extract immediate, evidence-based insights. This project proposes ClinicaRAG, an advanced Clinical Decision Support System (CDSS) that integrates domain-specific encoders with Retrieval-Augmented Generation (RAG) to provide verifiable diagnostic assistance. The system utilizes BioBERT or ClinicalBERT for fine-grained Named Entity Recognition (NER) to extract symptoms, diagnoses, and medications from raw clinical notes. To ensure medical accuracy and minimize hallucinations, the framework incorporates a RAG pipeline that fetches relevant medical evidence from trusted sources like PubMed and MedQuAD.

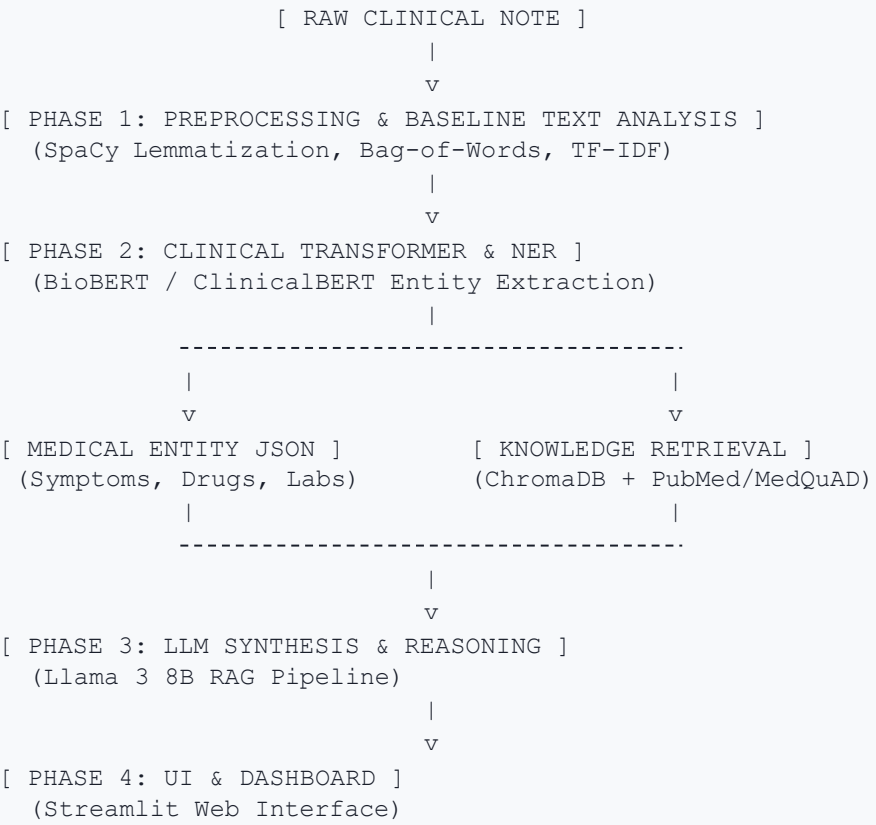
The proposed architecture involves rigorous text preprocessing using lemmatization and tokenization via SpaCy to clean and structure raw clinical text. A high-performance vector database, such as ChromaDB or FAISS, is implemented to index and search medical knowledge bases for context-aware evidence gathering. The final clinical synthesis and reasoning are performed by a quantized Llama 3 (8B) model, which generates suggested differential diagnoses, recommended tests, and treatment rationales with verifiable back-referenced citations.

The system's performance is evaluated using Precision, Recall, and F1-Score for the entity extraction module, alongside Hit Rate and Mean Reciprocal Rank (MRR) for the retrieval component. The resulting framework provides a transparent, Streamlit-based dashboard for healthcare professionals to interact with AI-driven clinical insights. By combining transformer-based extraction with generative reasoning, ClinicaRAG aims to provide actionable and grounded insights that can support medical decision-making and improve patient outcomes.

Datasets

Dataset	URL link
MedQuAD	https://github.com/abachaa/MedQuAD
PubMed / PubMedQA	https://huggingface.co/datasets/pubmed_qa
MIMIC-III	https://physionet.org/content/mimiciii/
Clinical Notes (Fallback)	https://huggingface.co/datasets/MTAI/clinical-notes-dataset

Approach



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